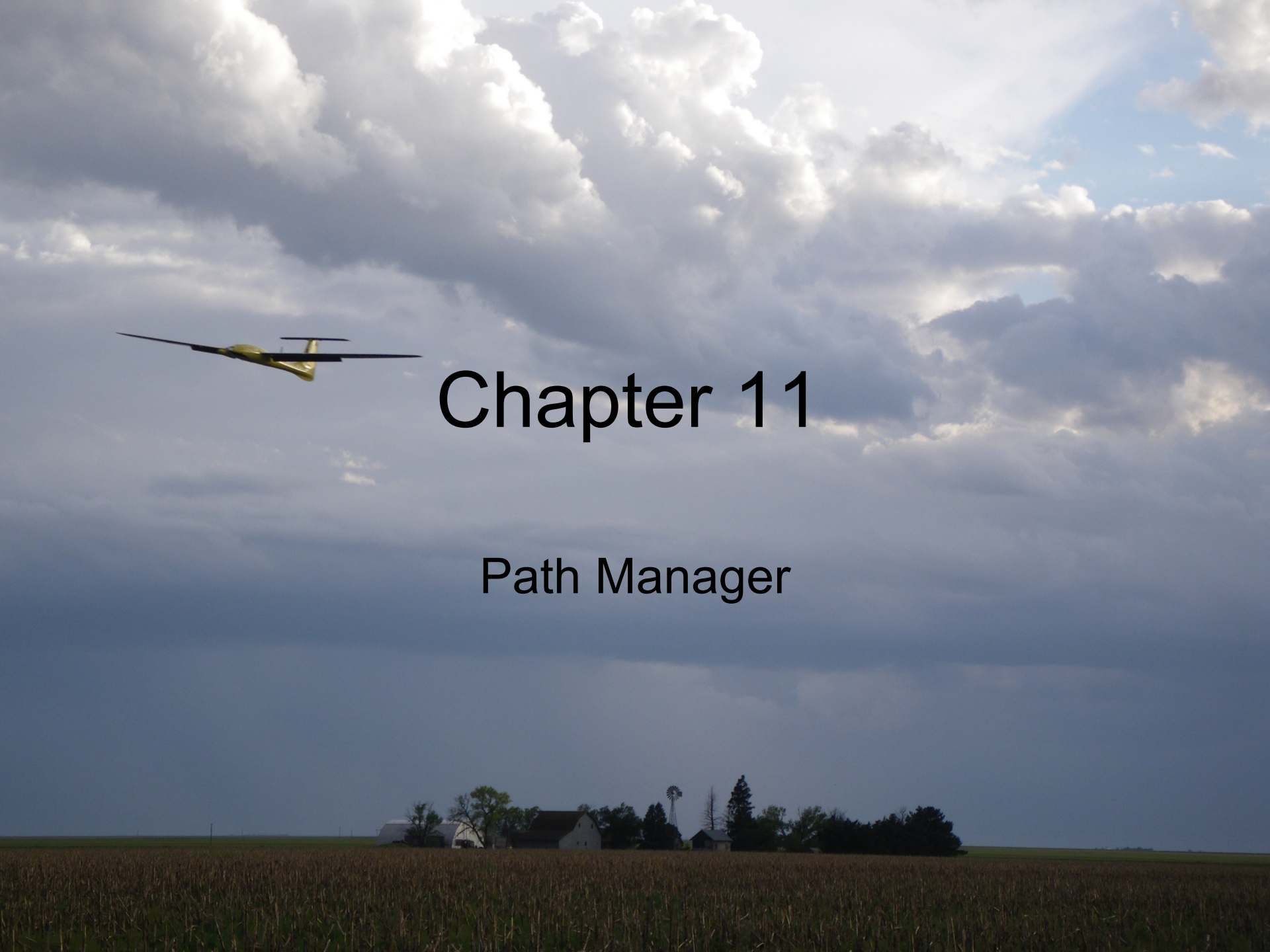
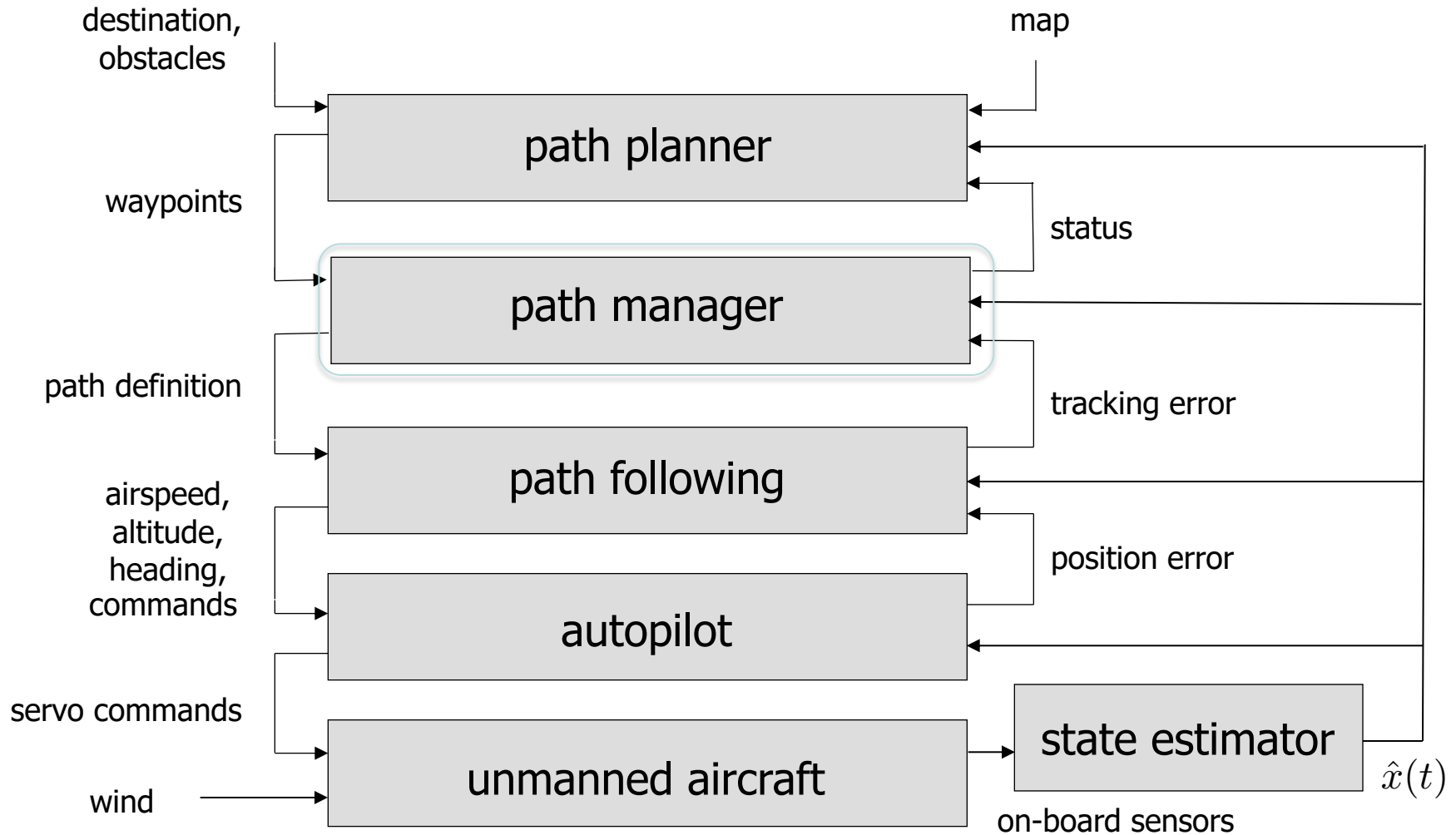


A yellow and black glider is shown in flight, moving from left to right across the upper left portion of the frame. The background is a vast, open landscape featuring a field of tall, golden-brown crops in the foreground. In the distance, a small farmstead is visible, including a white barn, a smaller house, and a windmill. The sky is filled with large, billowing white and grey clouds, with sunlight breaking through in several places, creating a dramatic and atmospheric scene.

Chapter 11

Path Manager

Control Architecture

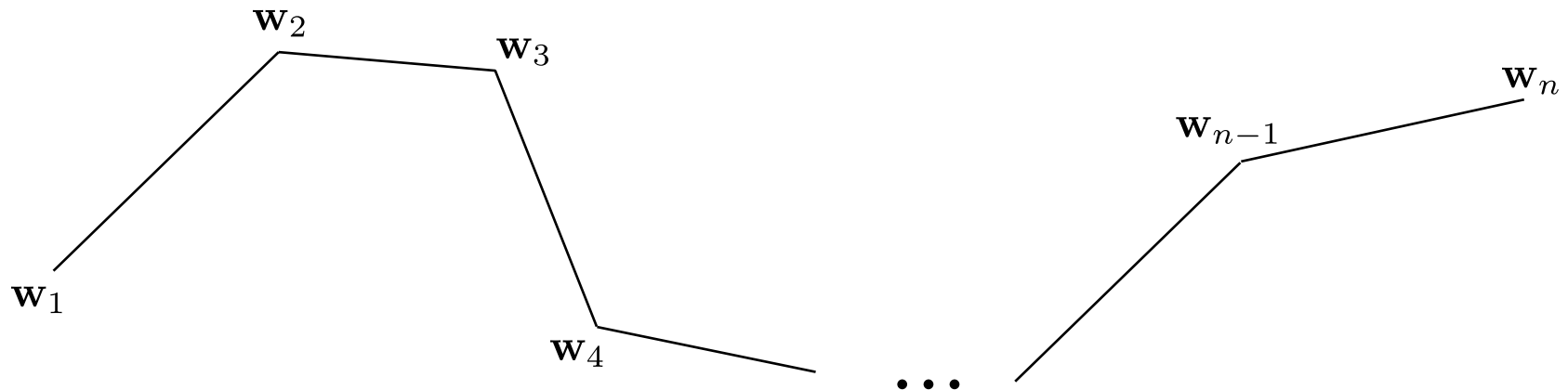


Path Definition

Waypoint path defined as ordered sequence of waypoints

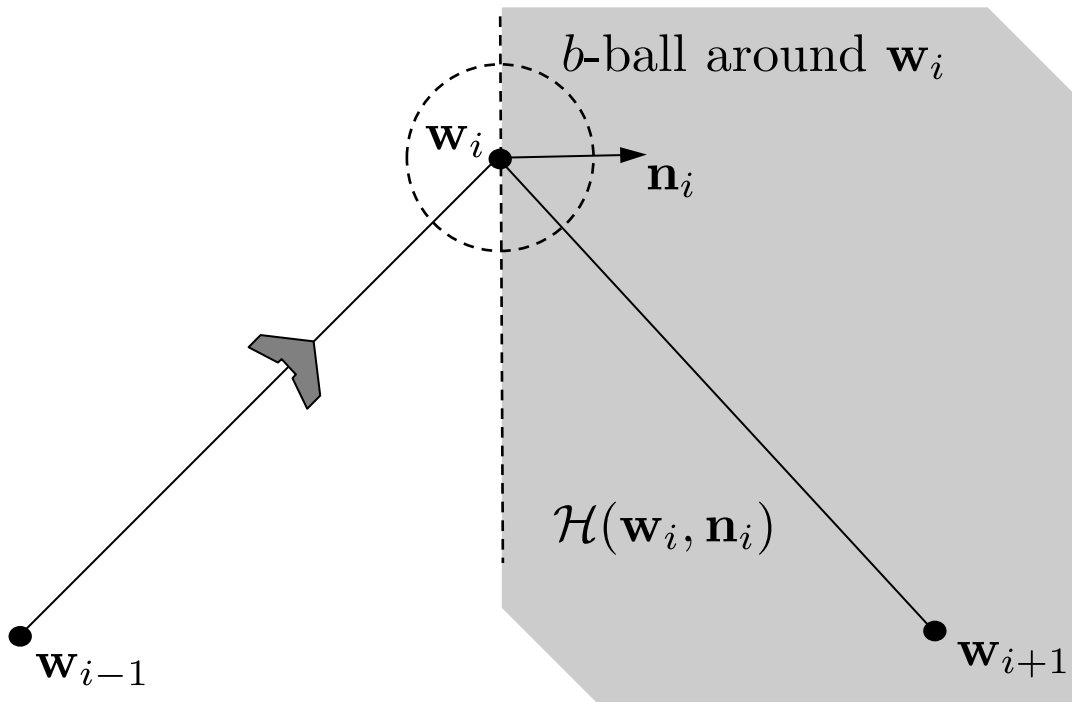
$$\mathcal{W} = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_N\}$$

where $\mathbf{w}_i = (w_{n,i}, w_{e,i}, w_{d,i})^\top \in \mathbb{R}^3$.

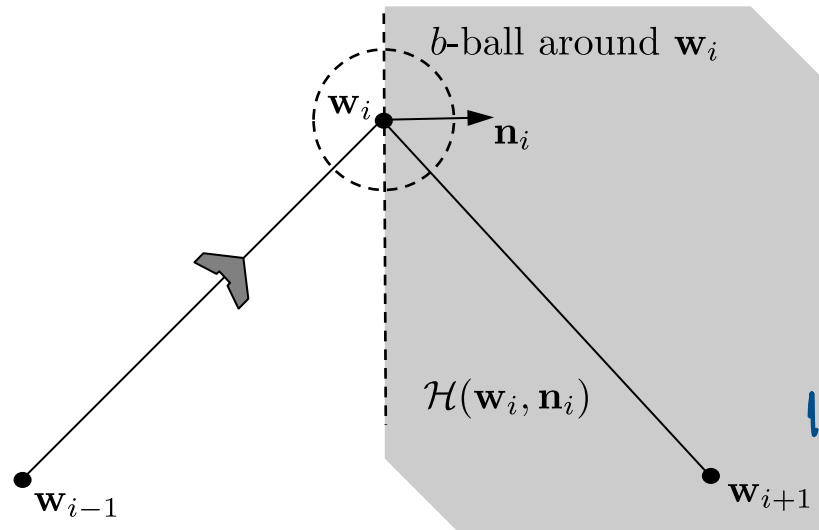


Waypoint Switching

- Two methods
 - b -ball around waypoint
 - half plane through waypoint



Waypoint Switching



Given point $\mathbf{r} \in \mathbb{R}^3$ and normal vector $\mathbf{n} \in \mathbb{R}^3$, define half plane

$$\mathcal{H}(\mathbf{r}, \mathbf{n}) \triangleq \{\mathbf{p} \in \mathbb{R}^3 : (\mathbf{p} - \mathbf{r})^\top \mathbf{n} \geq 0\}$$

Define unit vector pointing in direction of line $\overline{\mathbf{w}_i \mathbf{w}_{i+1}}$ as

$$\mathbf{q}_i \triangleq \frac{\mathbf{w}_{i+1} - \mathbf{w}_i}{\|\mathbf{w}_{i+1} - \mathbf{w}_i\|}$$

Unit normal to the 3-D half plane that separates the line $\overline{\mathbf{w}_{i-1} \mathbf{w}_i}$ from the line $\overline{\mathbf{w}_i \mathbf{w}_{i+1}}$ is given by

$$\mathbf{n}_i \triangleq \frac{\mathbf{q}_{i-1} + \mathbf{q}_i}{\|\mathbf{q}_{i-1} + \mathbf{q}_i\|}$$

MAV tracks straight-line path from $\mathbf{w}_{i-1}$ to $\mathbf{w}_i$ until it enters $\mathcal{H}(\mathbf{w}_i, \mathbf{n}_i)$, at which point it will track straight-line path from $\mathbf{w}_i$ to $\mathbf{w}_{i+1}$

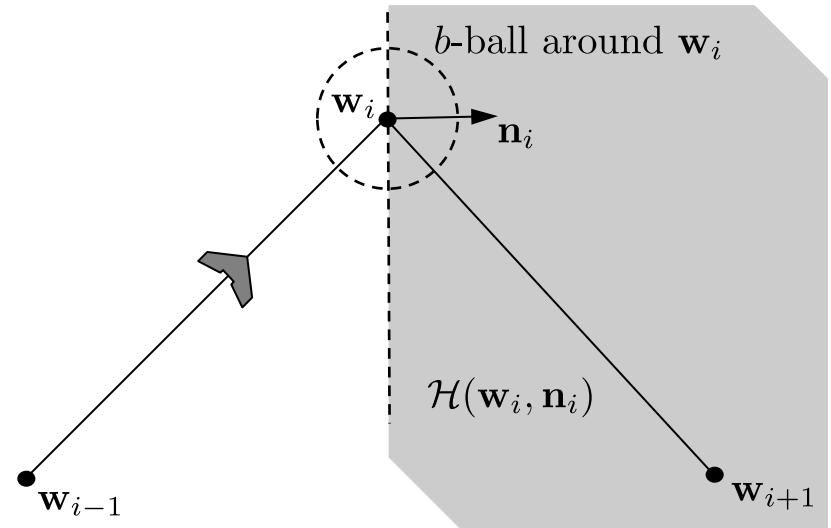
Waypoint Following

Algorithm 5 Follow Waypoints: $(\mathbf{r}, \mathbf{q}) = \text{followWpp}(\mathcal{W}, \mathbf{p})$

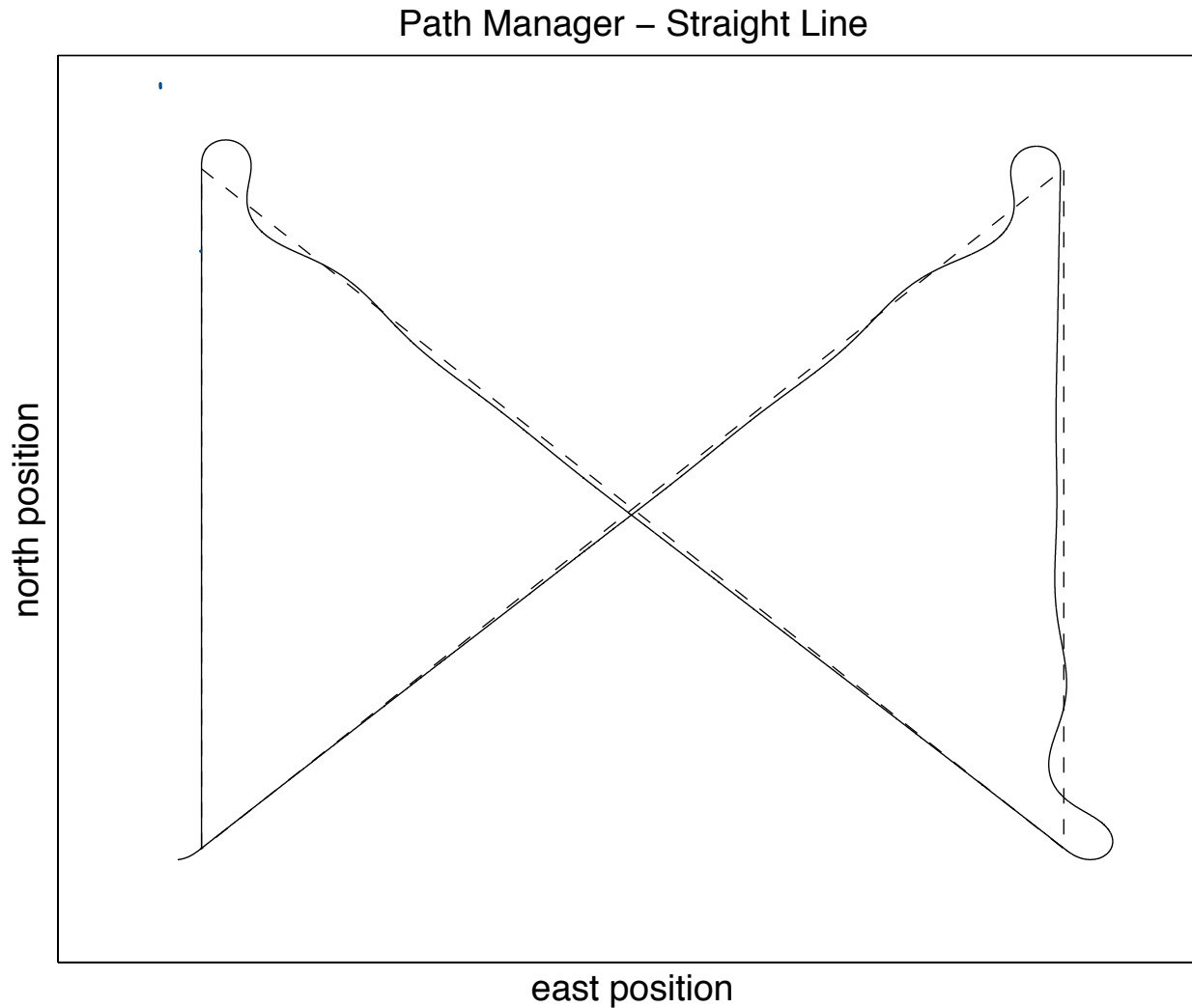
Input: Waypoint path $\mathcal{W} = \{\mathbf{w}_1, \dots, \mathbf{w}_N\}$, MAV position
 $\mathbf{p} = (p_n, p_e, p_d)^\top$.

Require: $N \geq 3$

- 1: **if** New waypoint path $\mathcal{W}$ is received **then**
 - 2: Initialize waypoint index: $i \leftarrow 2$
 - 3: **end if**
 - 4: $\mathbf{r} \leftarrow \mathbf{w}_{i-1}$
 - 5: $\mathbf{q}_{i-1} \leftarrow \frac{\mathbf{w}_i - \mathbf{w}_{i-1}}{\|\mathbf{w}_i - \mathbf{w}_{i-1}\|}$
 - 6: $\mathbf{q}_i \leftarrow \frac{\mathbf{w}_{i+1} - \mathbf{w}_i}{\|\mathbf{w}_{i+1} - \mathbf{w}_i\|}$
 - 7: $\mathbf{n}_i \leftarrow \frac{\mathbf{q}_{i-1} + \mathbf{q}_i}{\|\mathbf{q}_{i-1} + \mathbf{q}_i\|}$
 - 8: **if** $\mathbf{p} \in \mathcal{H}(\mathbf{w}_i, \mathbf{n}_i)$ **then**
 - 9: Increment $i \leftarrow (i + 1)$ until $i = N - 1$
 - 10: **end if**
 - 11: **return** $\mathbf{r}, \mathbf{q} = \mathbf{q}_{i-1}$ at each time step
-

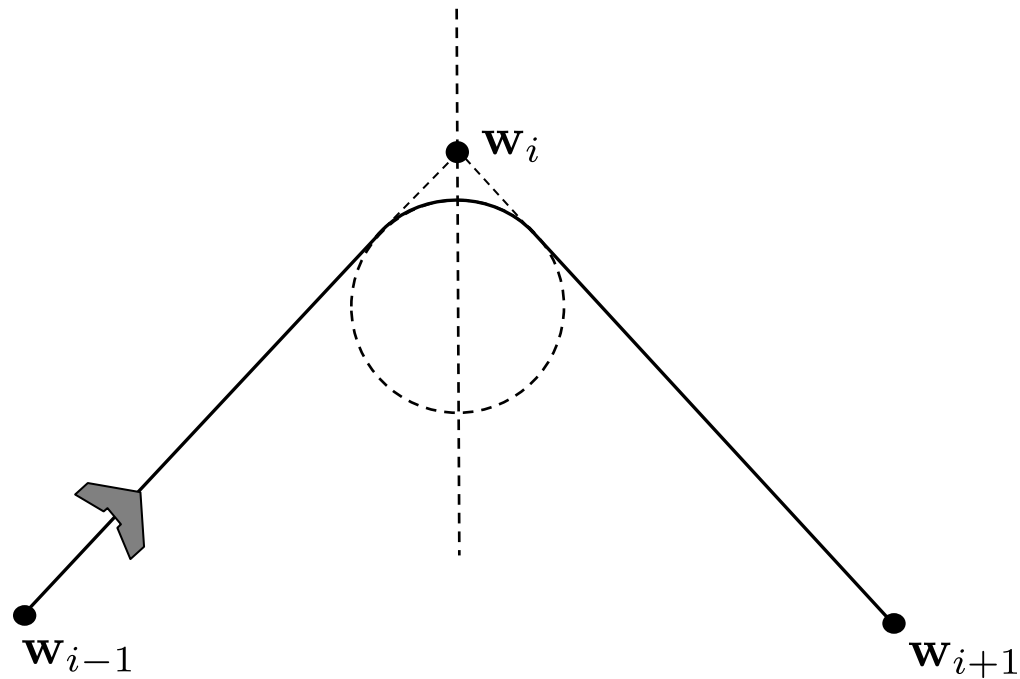


Waypoint Following Results

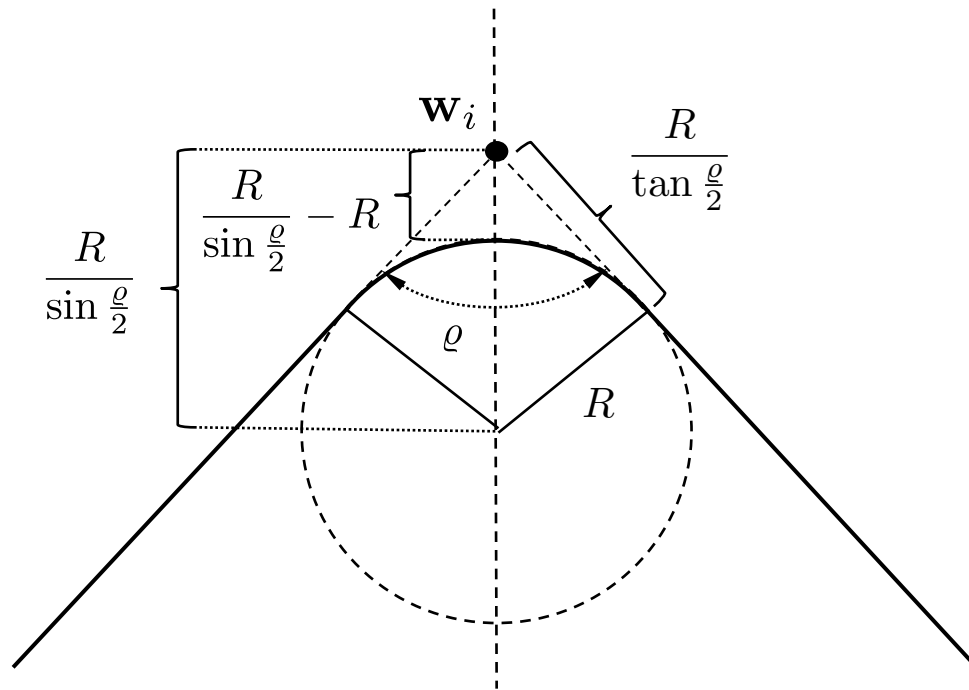


Fillet Transition

Transition between path segments can be smoothed by adding a fillet



Fillet Geometry



Fillet Smoothing Half Planes

Fillet center defined as

$$\mathbf{c} = \mathbf{w}_i - \left(\frac{R}{\sin \frac{\varrho}{2}} \right) \frac{\mathbf{q}_{i-1} - \mathbf{q}_i}{\|\mathbf{q}_{i-1} - \mathbf{q}_i\|}$$

Half plane $\mathcal{H}_1$ is defined by location

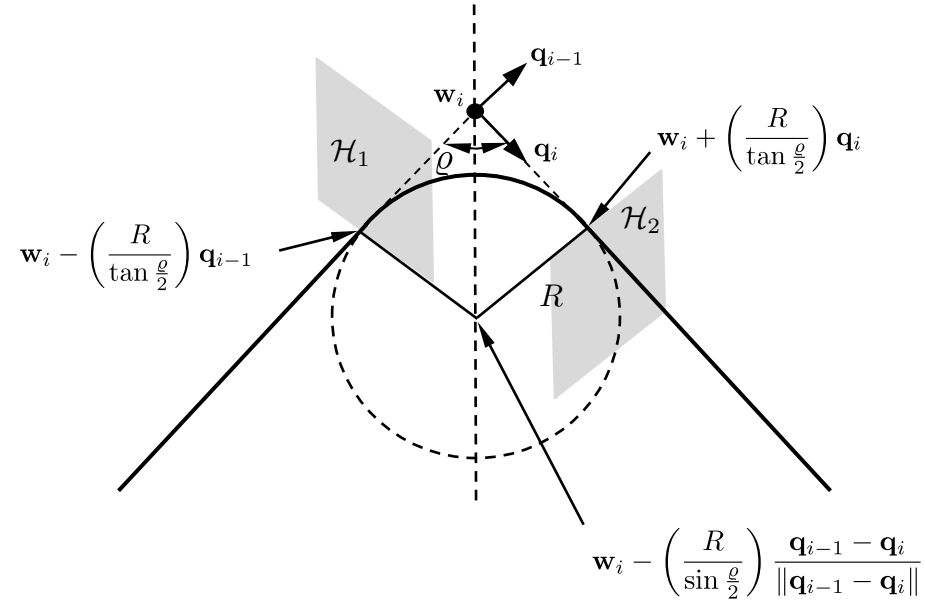
$$\mathbf{r}_1 = \mathbf{w}_i - \left(\frac{R}{\tan \frac{\varrho}{2}} \right) \mathbf{q}_{i-1}$$

and normal vector $\mathbf{q}_{i-1}$

Half plane $\mathcal{H}_2$ is defined by location

$$\mathbf{r}_2 = \mathbf{w}_i + \left(\frac{R}{\tan \frac{\varrho}{2}} \right) \mathbf{q}_i$$

and normal vector $\mathbf{q}_i$



Waypoint Following with Fillets

Algorithm 1 Follow Waypoints with Fillets: $(\text{flag}, \mathbf{r}, \mathbf{q}, \mathbf{c}, \rho, \lambda) = \text{followWppFillet}(\mathcal{W}, \mathbf{p}, R)$

Ensure: Waypoint path $\mathcal{W} = \{\mathbf{w}_1, \dots, \mathbf{w}_N\}$, MAV position $\mathbf{p} = (p_n, p_e, p_d)^\top$,
fillet radius R .

Require: $N \geq 3$.

1: **if** New waypoint path $\mathcal{W}$ is received **then**

2: Initialize waypoint index: $i \leftarrow 2$, and state machine: $\text{state} \leftarrow 1$.

3: **end if**

4: $\mathbf{q}_{i-1} \leftarrow \frac{\mathbf{w}_i - \mathbf{w}_{i-1}}{\|\mathbf{w}_i - \mathbf{w}_{i-1}\|}$.

5: $\mathbf{q}_i \leftarrow \frac{\mathbf{w}_{i+1} - \mathbf{w}_i}{\|\mathbf{w}_{i+1} - \mathbf{w}_i\|}$.

6: $\varrho \leftarrow \cos^{-1}(-\mathbf{q}_{i-1}^\top \mathbf{q}_i)$.

7: **if** $\text{state} = 1$ **then**

8: $\text{flag} \leftarrow 1$

9: $\mathbf{r} \leftarrow \mathbf{w}_{i-1}$

10: $\mathbf{q} \leftarrow \mathbf{q}_{i-1}$

11: $\mathbf{z} \leftarrow \mathbf{w}_i - \left(\frac{R}{\tan(\varrho/2)}\right) \mathbf{q}_{i-1}$

12: **if** $\mathbf{p} \in \mathcal{H}(\mathbf{z}, \mathbf{q}_{i-1})$ **then**

13: $\text{state} \leftarrow 2$

14: **end if**

15: **else if** $\text{state} = 2$ **then**

16: $\text{flag} \leftarrow 2$

17: $\mathbf{c} \leftarrow \mathbf{w}_i - \left(\frac{R}{\sin(\varrho/2)}\right) \frac{\mathbf{q}_{i-1} - \mathbf{q}_i}{\|\mathbf{q}_{i-1} - \mathbf{q}_i\|}$

18: $\rho \leftarrow R$

19: $\lambda \leftarrow \text{sign}(q_{i-1,n}q_{i,e} - q_{i-1,e}q_{i,n})$.

20: $\mathbf{z} \leftarrow \mathbf{w}_i + \left(\frac{R}{\tan(\varrho/2)}\right) \mathbf{q}_i$

21: **if** $\mathbf{p} \in \mathcal{H}(\mathbf{z}, \mathbf{q}_i)$ **then**

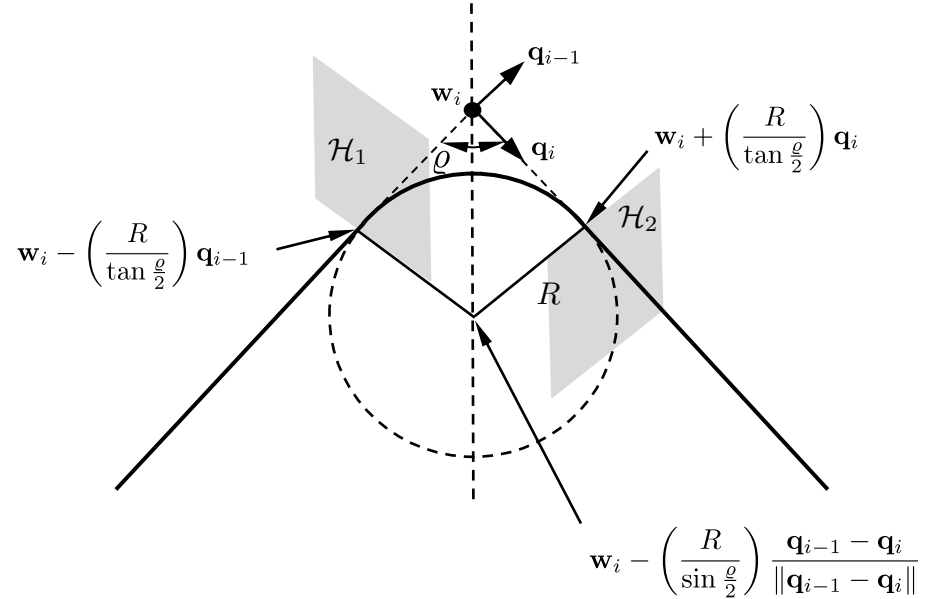
22: $i \leftarrow (i+1)$ until $i = N-1$.

23: $\text{state} \leftarrow 1$

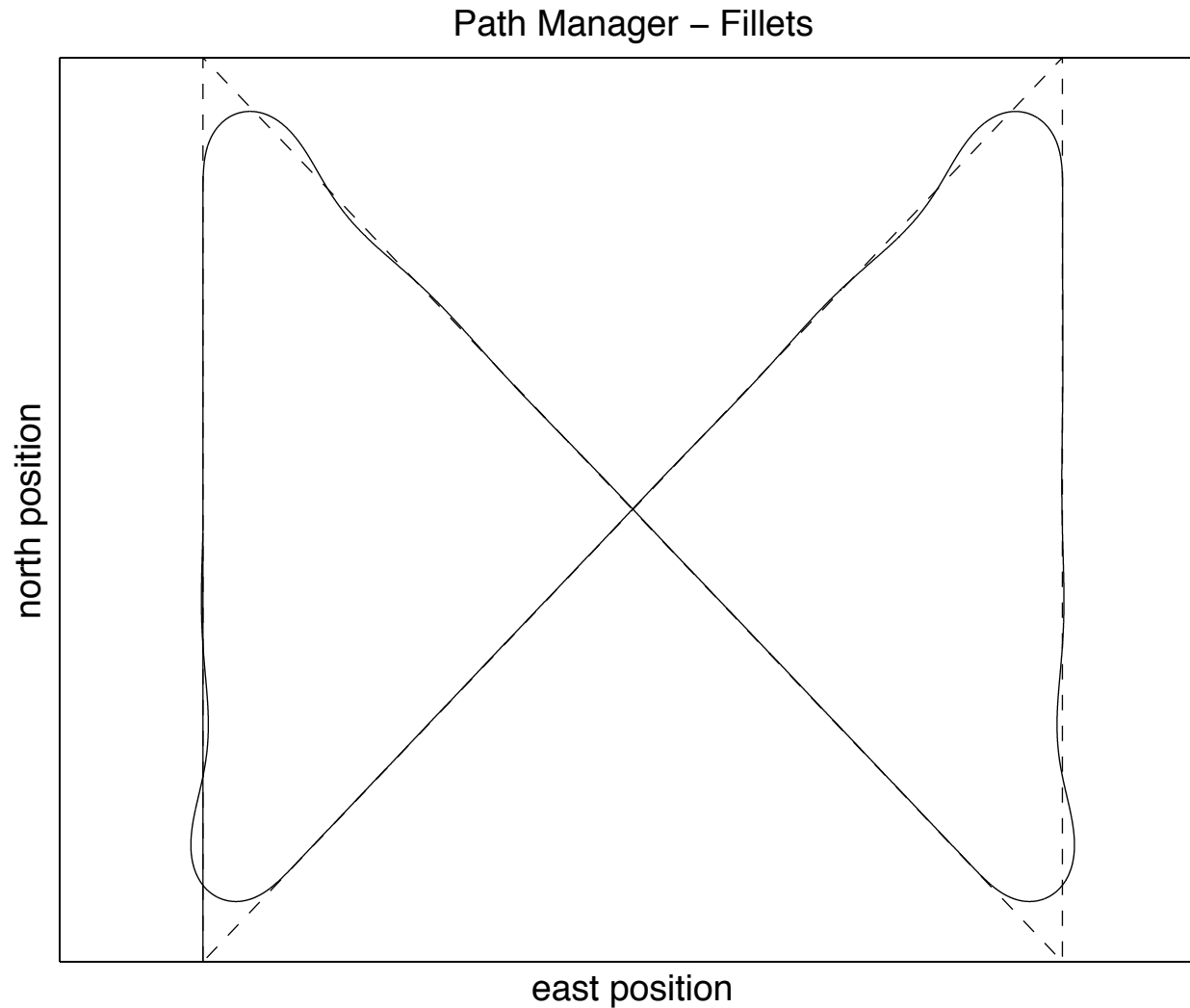
24: **end if**

25: **end if**

26: **return** $\text{flag}, \mathbf{r}, \mathbf{q}, \mathbf{c}, \rho, \lambda$.



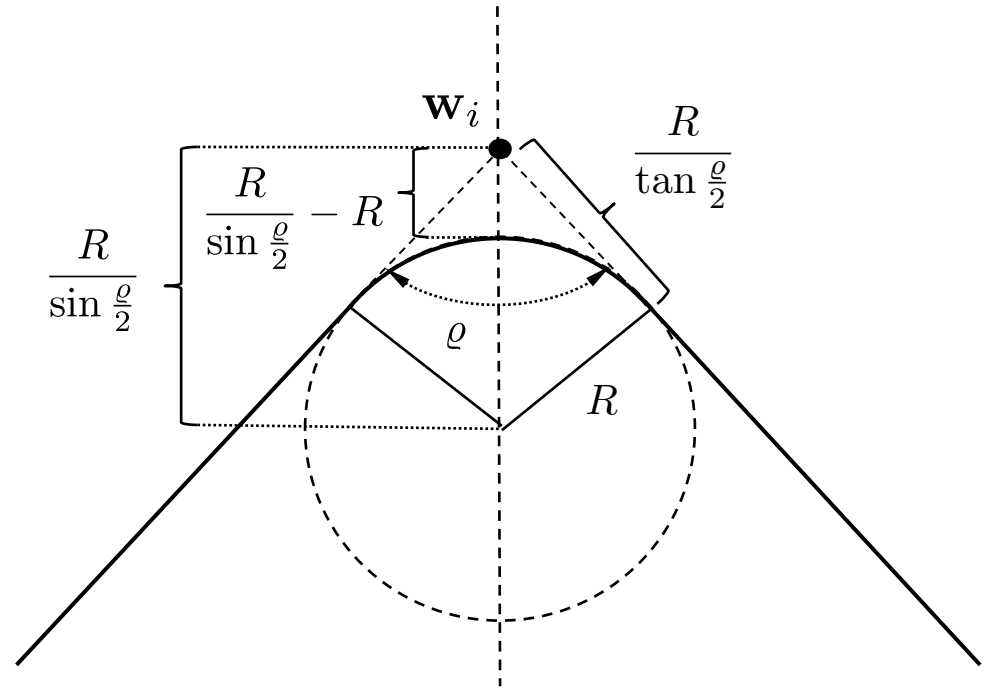
Waypoint Following with Fillets



Fillet Path Length

Straight-line path length (no fillets):

$$|\mathcal{W}| \triangleq \sum_{i=2}^N \|\mathbf{w}_i - \mathbf{w}_{i-1}\|.$$



Path length with fillets:

$$|\mathcal{W}|_F = |\mathcal{W}| + \sum_{i=2}^N \left(R \varrho_i - \frac{2R}{\tan \frac{\varrho_i}{2}} \right).$$

Dubins Paths

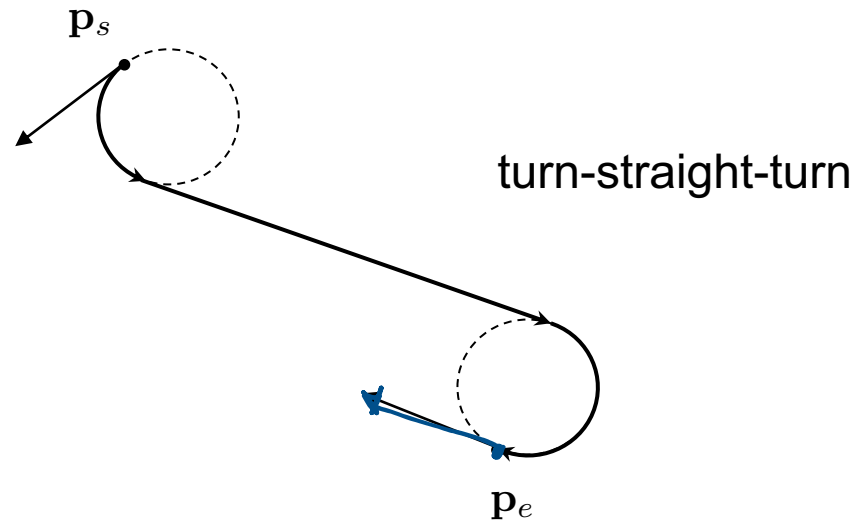
For vehicle with kinematics given by

$$\dot{p}_n = V \cos \vartheta$$

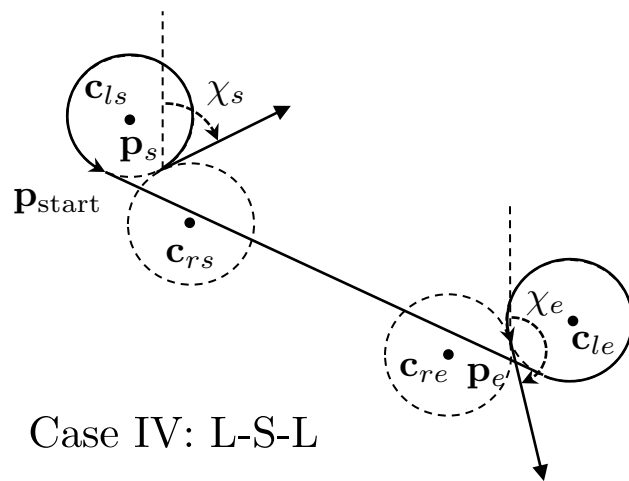
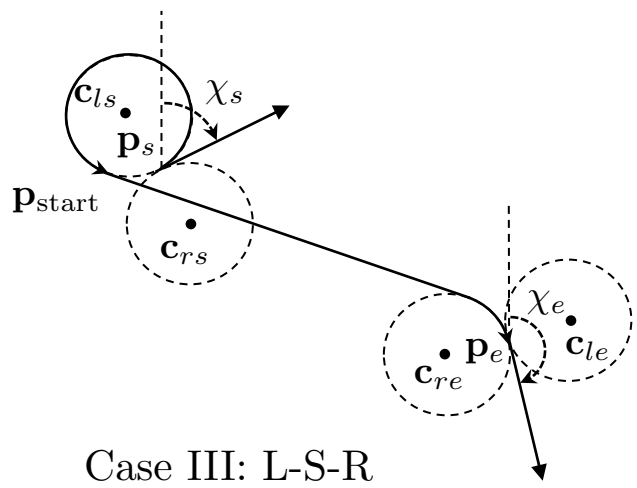
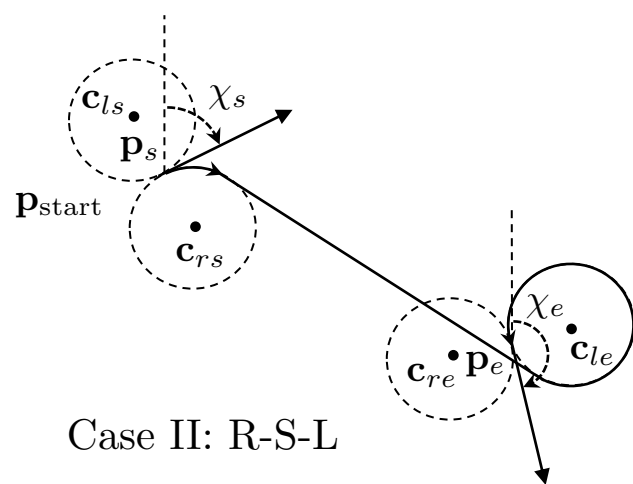
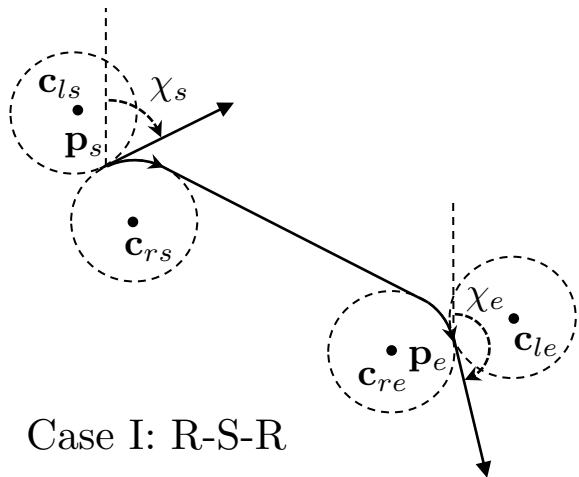
$$\dot{p}_e = V \sin \vartheta$$

$$\dot{\vartheta} = u,$$

where V is constant and $u \in [-\bar{u}, \bar{u}]$, time-optimal path between two different configurations consists of circular arc, followed by straight line, and concluding with another circular arc to the final configuration, where the radius of the circular arcs is $V/\bar{u}$.



Four Cases



Dubins path is defined as shortest of four cases

Path Length Preliminaries

Given position $\mathbf{p}$, course χ , and radius R , centers of right and left turning circles are given by

$$\mathbf{c}_r = \mathbf{p} + R \left(\cos\left(\chi + \frac{\pi}{2}\right), \sin\left(\chi + \frac{\pi}{2}\right), 0 \right)^\top$$

$$\mathbf{c}_l = \mathbf{p} + R \left(\cos\left(\chi - \frac{\pi}{2}\right), \sin\left(\chi - \frac{\pi}{2}\right), 0 \right)^\top$$

For clockwise circles, angular distance between ϑ_1 and ϑ_2 given by $\langle 2\pi + \vartheta_2 - \vartheta_1 \rangle$

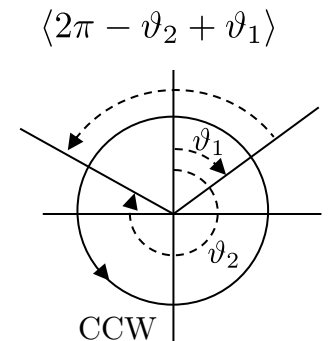
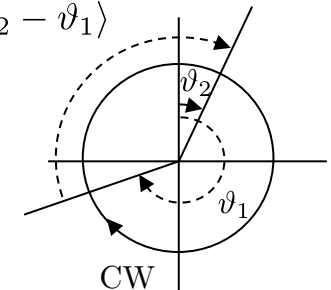
$$|\vartheta_2 - \vartheta_1|_{CW} \triangleq \langle 2\pi + \vartheta_2 - \vartheta_1 \rangle,$$

where

$$\langle \varphi \rangle \triangleq \varphi \mod 2\pi$$

For counter clockwise circles,

$$|\vartheta_2 - \vartheta_1|_{CCW} \triangleq \langle 2\pi - \vartheta_2 + \vartheta_1 \rangle$$



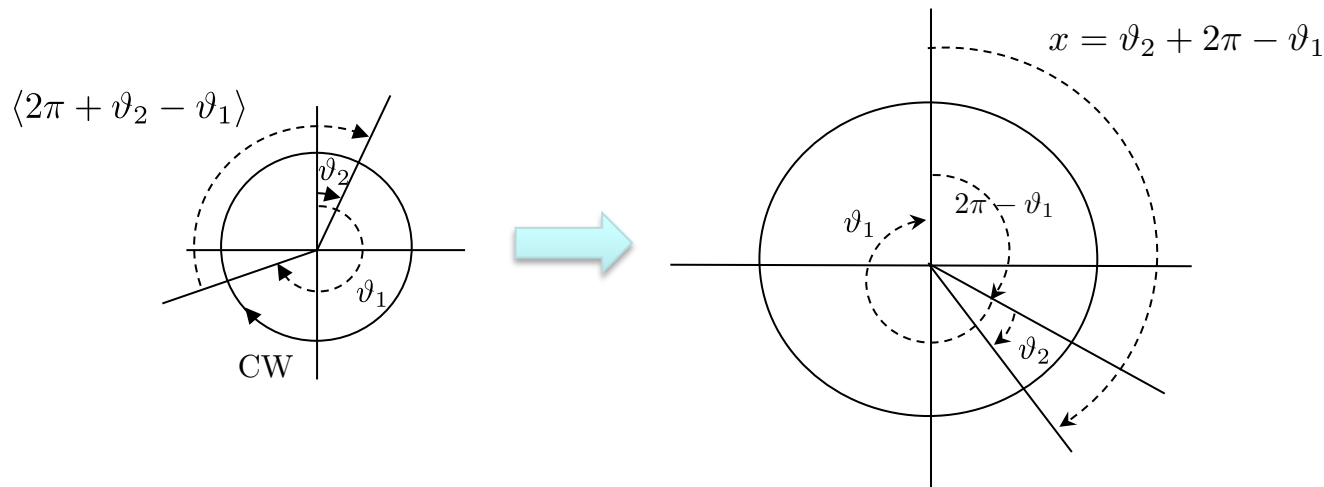
Alternative Viewpoint

For clockwise circles, angular distance between ϑ_1 and ϑ_2 given by

$$|\vartheta_2 - \vartheta_1|_{CW} \triangleq \langle 2\pi + \vartheta_2 - \vartheta_1 \rangle,$$

where

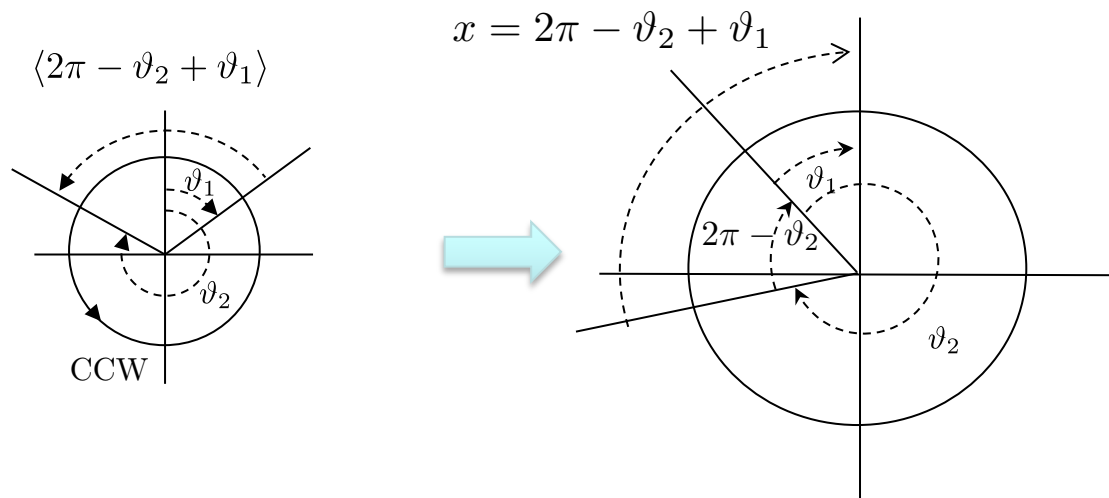
$$\langle \varphi \rangle \triangleq \varphi \mod 2\pi$$



Alternative Viewpoint

For counter clockwise circles,

$$|\vartheta_2 - \vartheta_1|_{CCW} \triangleq \langle 2\pi - \vartheta_2 + \vartheta_1 \rangle$$



Dubins Case I: R-S-R

Distance traveled along $\mathbf{c}_{rs}$

$$R\langle 2\pi + \langle \vartheta - \frac{\pi}{2} \rangle - \langle \chi_s - \frac{\pi}{2} \rangle \rangle$$

Distance traveled along $\mathbf{c}_{re}$

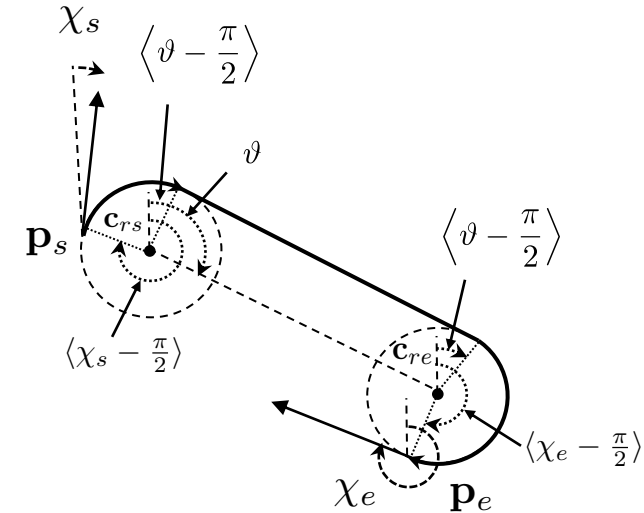
$$R\langle 2\pi + \langle \chi_e - \frac{\pi}{2} \rangle - \langle \vartheta - \frac{\pi}{2} \rangle \rangle$$

where

$$\vartheta = \text{atan2}(c_{ree} - c_{rse}, c_{ren} - c_{rsn})$$

Total path length:

$$L_1 = \|\mathbf{c}_{rs} - \mathbf{c}_{re}\| + R\langle 2\pi + \langle \vartheta - \frac{\pi}{2} \rangle - \langle \chi_s - \frac{\pi}{2} \rangle \rangle + R\langle 2\pi + \langle \chi_e - \frac{\pi}{2} \rangle - \langle \vartheta - \frac{\pi}{2} \rangle \rangle$$



Dubins Case II: R-S-L

Distance traveled along $\mathbf{c}_{rs}$

$$R\langle 2\pi + \langle \vartheta_2 \rangle - \langle \chi_s - \frac{\pi}{2} \rangle \rangle$$

Distance traveled along $\mathbf{c}_{le}$

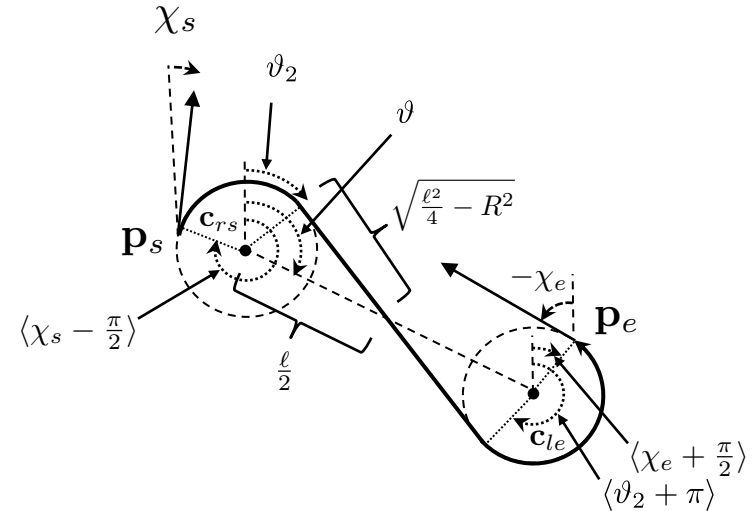
$$R\langle 2\pi + \langle \vartheta_2 + \pi \rangle - \langle \chi_e + \frac{\pi}{2} \rangle \rangle$$

where

$$\vartheta_2 = \vartheta - \frac{\pi}{2} + \sin^{-1} \left(\frac{2R}{\ell} \right)$$

Total path length:

$$L_2 = \sqrt{\ell^2 - 4R^2} + R\langle 2\pi + \langle \vartheta_2 \rangle - \langle \chi_s - \frac{\pi}{2} \rangle \rangle + R\langle 2\pi + \langle \vartheta_2 + \pi \rangle - \langle \chi_e + \frac{\pi}{2} \rangle \rangle$$



Dubins Case III: L-S-R

Distance traveled along $\mathbf{c}_{ls}$

$$R\langle 2\pi + \langle \chi_s + \frac{\pi}{2} \rangle - \langle \vartheta + \vartheta_2 \rangle \rangle$$

Distance traveled along $\mathbf{c}_{re}$

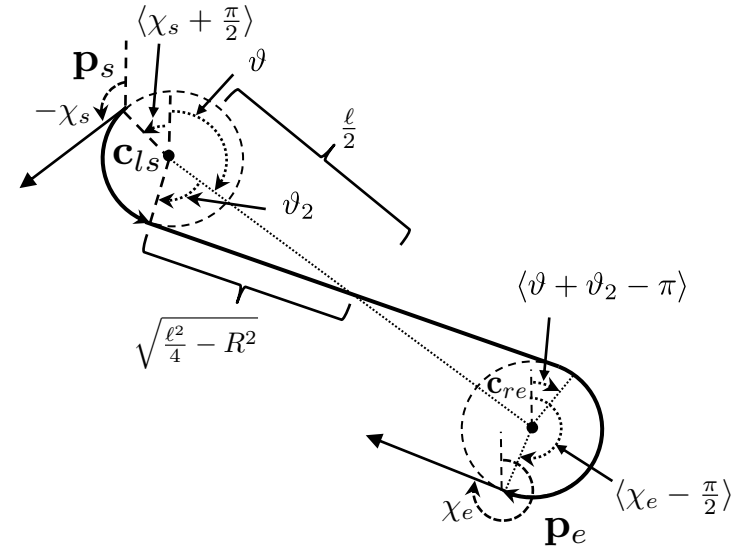
$$R\langle 2\pi + \langle \chi_e - \frac{\pi}{2} \rangle - \langle \vartheta + \vartheta_2 - \pi \rangle \rangle$$

where

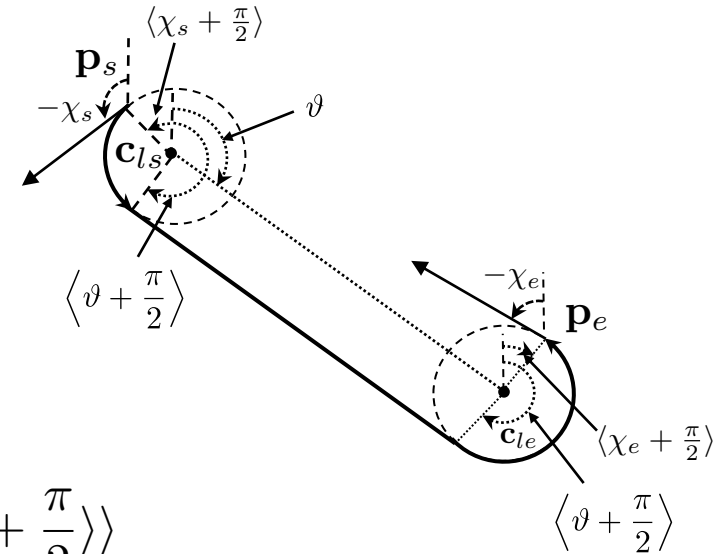
$$\vartheta_2 = \cos^{-1} \left(\frac{2R}{\ell} \right)$$

Total path length:

$$L_3 = \sqrt{\ell^2 - 4R^2} + R\langle 2\pi + \langle \chi_s + \frac{\pi}{2} \rangle - \langle \vartheta + \vartheta_2 \rangle \rangle + R\langle 2\pi + \langle \chi_e - \frac{\pi}{2} \rangle - \langle \vartheta + \vartheta_2 - \pi \rangle \rangle$$



Dubins Case IV: L-S-L



Distance traveled along $\mathbf{c}_{ls}$

$$R\langle 2\pi + \langle \chi_s + \frac{\pi}{2} \rangle - \langle \vartheta + \frac{\pi}{2} \rangle \rangle$$

Distance traveled along $\mathbf{c}_{le}$

$$R\langle 2\pi + \langle \vartheta + \frac{\pi}{2} \rangle - \langle \chi_e + \frac{\pi}{2} \rangle \rangle$$

Total path length:

$$L_4 = \|\mathbf{c}_{ls} - \mathbf{c}_{le}\| + R\langle 2\pi + \langle \chi_s + \frac{\pi}{2} \rangle - \langle \vartheta + \frac{\pi}{2} \rangle \rangle + R\langle 2\pi + \langle \vartheta + \frac{\pi}{2} \rangle - \langle \chi_e + \frac{\pi}{2} \rangle \rangle$$

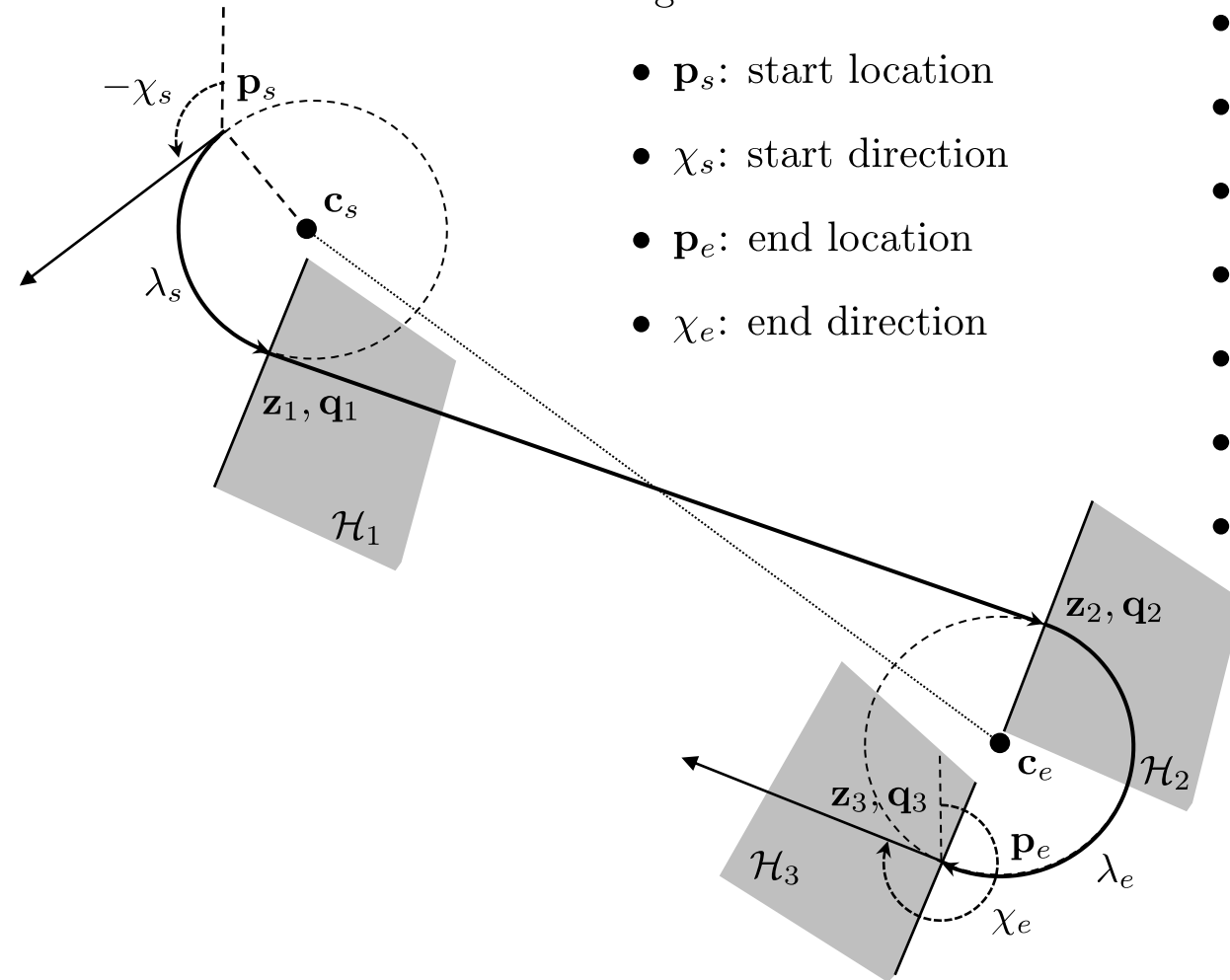
Dubins Path Half-plane Switching

Given desired start and end configurations

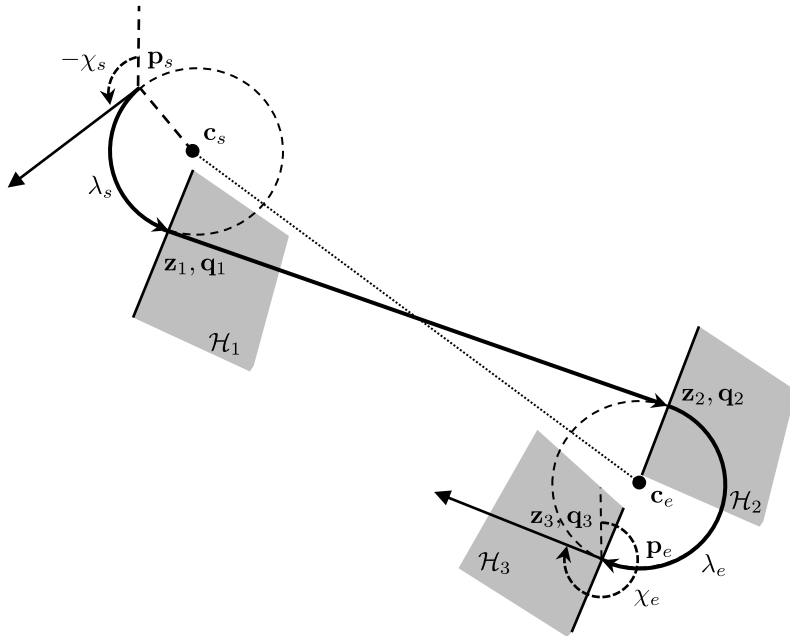
- $\mathbf{p}_s$: start location
- χ_s : start direction
- $\mathbf{p}_e$: end location
- χ_e : end direction

Calculate Dubins path parameters

- $\mathbf{c}_s$: start circle location
- λ_s : start circle direction
- $\mathbf{c}_e$: end circle location
- λ_e : end circle direction
- $\mathbf{z}_1, \mathbf{q}_1$: half-plane $\mathcal{H}_1$ parameters
- $\mathbf{z}_2, \mathbf{q}_2$: half-plane $\mathcal{H}_2$ parameters
- $\mathbf{z}_3, \mathbf{q}_3$: half-plane $\mathcal{H}_3$ parameters



Dubins Path Parameters Algorithm



Algorithm 7 Find Dubins Parameters:

$(L, c_s, \lambda_s, c_e, \lambda_e, z_1, q_1, z_2, z_3, q_3) =$
 findDubinsParameters($p_s, \chi_s, p_e, \chi_e, R$)

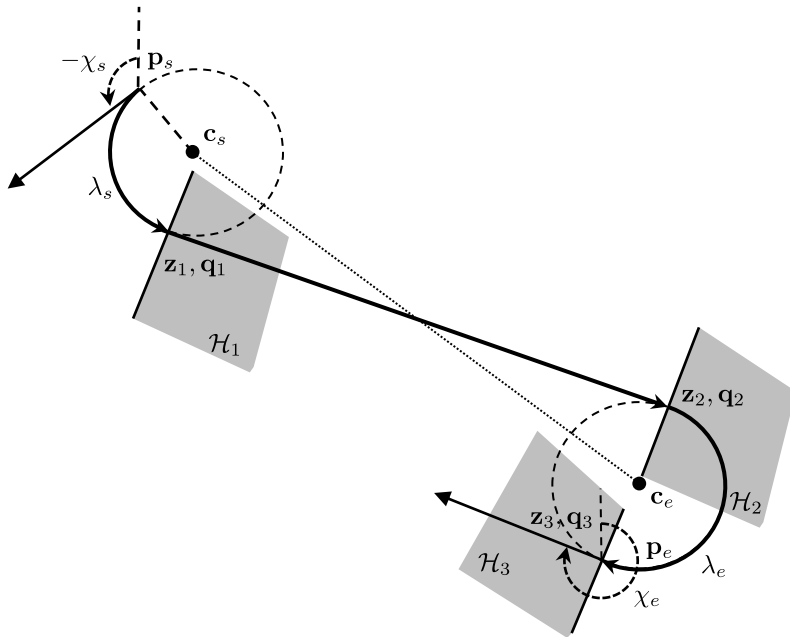
Input: Start configuration (p_s, χ_s), End configuration (p_e, χ_e),
 Radius R .

Require: $\|p_s - p_e\| \geq 3R$

Require: R is larger than minimum turn radius of MAV

- 1: $c_{rs} \leftarrow p_s + R\mathcal{R}_z\left(\frac{\pi}{2}\right)(\cos \chi_s, \sin \chi_s, 0)^\top$
- 2: $c_{ls} \leftarrow p_s + R\mathcal{R}_z\left(-\frac{\pi}{2}\right)(\cos \chi_s, \sin \chi_s, 0)^\top$
- 3: $c_{re} \leftarrow p_e + R\mathcal{R}_z\left(\frac{\pi}{2}\right)(\cos \chi_e, \sin \chi_e, 0)^\top$
- 4: $c_{le} \leftarrow p_e + R\mathcal{R}_z\left(-\frac{\pi}{2}\right)(\cos \chi_e, \sin \chi_e, 0)^\top$
- 5: Compute L_1, L_2, L_3 , and L_4 using equations (11.9) through (11.12).
- 6: $L \leftarrow \min\{L_1, L_2, L_3, L_4\}$
- 7: **if** $\arg \min\{L_1, L_2, L_3, L_4\} = 1$ **then**
- 8: $c_s \leftarrow c_{rs}, \quad \lambda_s \leftarrow +1, \quad c_e \leftarrow c_{re}, \quad \lambda_e \leftarrow +1$
- 9: $q_1 \leftarrow \frac{c_e - c_s}{\|c_e - c_s\|}$
- 10: $z_1 \leftarrow c_s + R\mathcal{R}_z\left(-\frac{\pi}{2}\right)q_1$
- 11: $z_2 \leftarrow c_e + R\mathcal{R}_z\left(-\frac{\pi}{2}\right)q_1$
- 12: **else if** $\arg \min\{L_1, L_2, L_3, L_4\} = 2$ **then**
- 13: $c_s \leftarrow c_{rs}, \quad \lambda_s \leftarrow +1, \quad c_e \leftarrow c_{le}, \quad \lambda_e \leftarrow -1$
- 14: $\ell \leftarrow \|c_e - c_s\|$
- 15: $\vartheta \leftarrow \text{angle}(c_e - c_s)$
- 16: $\vartheta_2 \leftarrow \vartheta - \frac{\pi}{2} + \sin^{-1} \frac{2R}{\ell}$
- 17: $q_1 \leftarrow \mathcal{R}_z\left(\vartheta_2 + \frac{\pi}{2}\right)e_1$
- 18: $z_1 \leftarrow c_s + R\mathcal{R}_z(\vartheta_2)e_1$
- 19: $z_2 \leftarrow c_e + R\mathcal{R}_z(\vartheta_2 + \pi)e_1$
- 20: **else if** $\arg \min\{L_1, L_2, L_3, L_4\} = 3$ **then**
- 21: $c_s \leftarrow c_{ls}, \quad \lambda_s \leftarrow -1, \quad c_e \leftarrow c_{re}, \quad \lambda_e \leftarrow +1$
- 22: $\ell \leftarrow \|c_e - c_s\|$
- 23: $\vartheta \leftarrow \text{angle}(c_e - c_s)$
- 24: $\vartheta_2 \leftarrow \cos^{-1} \frac{2R}{\ell}$
- 25: $q_1 \leftarrow \mathcal{R}_z\left(\vartheta + \vartheta_2 - \frac{\pi}{2}\right)e_1$
- 26: $z_1 \leftarrow c_s + R\mathcal{R}_z(\vartheta + \vartheta_2)e_1$
- 27: $z_2 \leftarrow c_e + R\mathcal{R}_z(\vartheta + \vartheta_2 - \pi)e_1$
- 28: **else if** $\arg \min\{L_1, L_2, L_3, L_4\} = 4$ **then**
- 29: $c_s \leftarrow c_{ls}, \quad \lambda_s \leftarrow -1, \quad c_e \leftarrow c_{le}, \quad \lambda_e \leftarrow -1$
- 30: $q_1 \leftarrow \frac{c_e - c_s}{\|c_e - c_s\|}$
- 31: $z_1 \leftarrow c_s + R\mathcal{R}_z\left(\frac{\pi}{2}\right)q_1$
- 32: $z_2 \leftarrow c_e + R\mathcal{R}_z\left(\frac{\pi}{2}\right)q_2$
- 33: **end if**
- 34: $z_3 \leftarrow p_e$
- 35: $q_3 \leftarrow \mathcal{R}_z(\chi_e)e_1$

Dubins Path Following Algorithm



Algorithm 8 Follow Waypoints with Dubins: $(\text{flag}, \mathbf{r}, \mathbf{q}, \mathbf{c}, \rho, \lambda) = \text{followWppDubins}(\mathcal{P}, \mathbf{p}, R)$

Input: Configuration path $\mathcal{P} = \{(\mathbf{w}_1, \chi_1), \dots, (\mathbf{w}_N, \chi_N)\}$, MAV position $\mathbf{p} = (p_n, p_e, p_d)^\top$, fillet radius R .

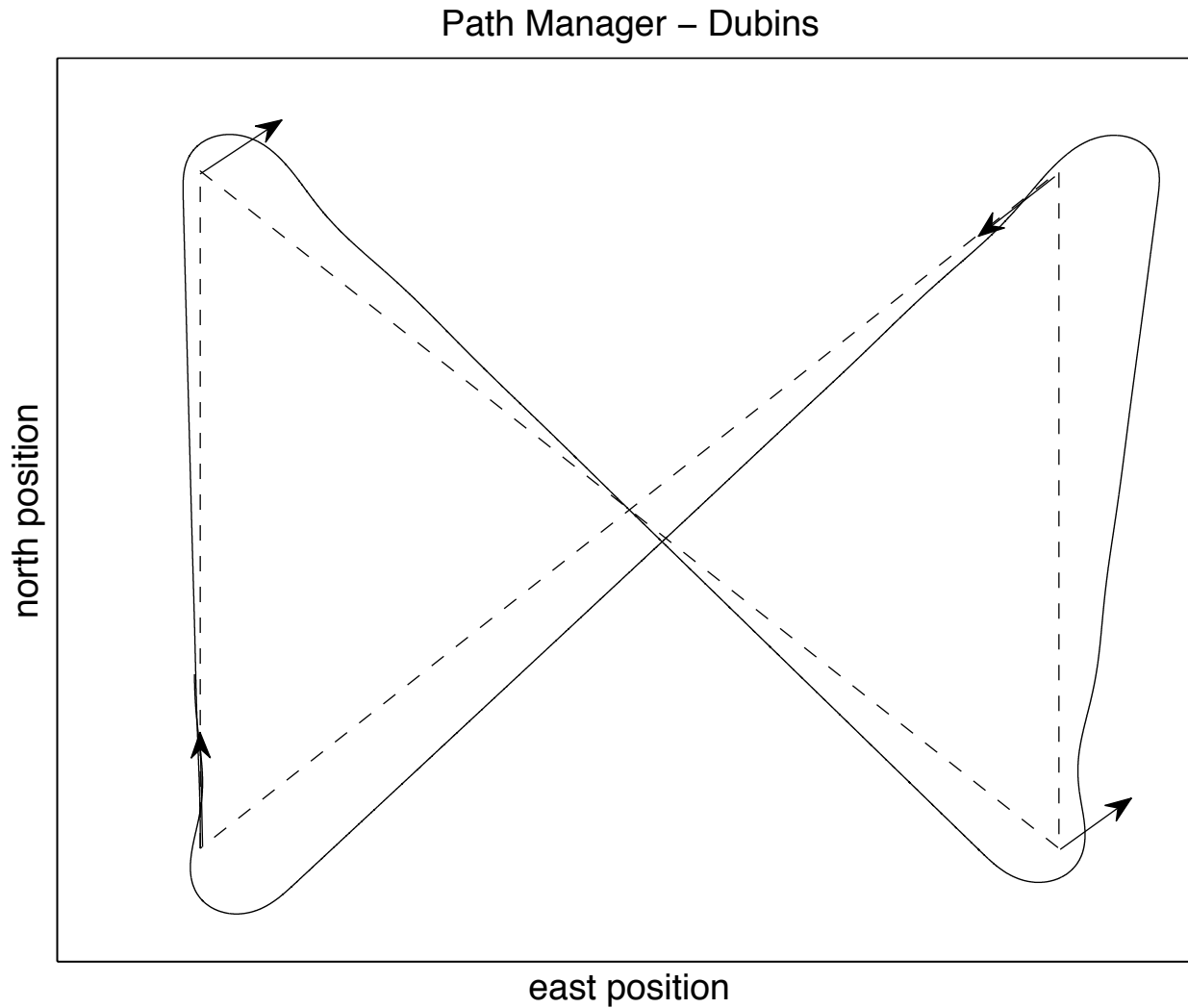
Require: $N \geq 3$

```

1: if New configuration path  $\mathcal{P}$  is received then
2:   Initialize waypoint pointer:  $i \leftarrow 2$ , and state machine: state  $\leftarrow 1$ .
3: end if
4:  $(L, \mathbf{c}_s, \lambda_s, \mathbf{c}_e, \lambda_e, \mathbf{z}_1, \mathbf{q}_1, \mathbf{z}_2, \mathbf{z}_3, \mathbf{q}_3) \leftarrow$ 
   findDubinsParameters( $\mathbf{w}_{i-1}, \chi_{i-1}, \mathbf{w}_i, \chi_i, R$ )
5: if state = 1 then
6:   flag  $\leftarrow 2, \mathbf{c} \leftarrow \mathbf{c}_s, \rho \leftarrow R, \lambda \leftarrow \lambda_s$ 
7:   if  $\mathbf{p} \in \mathcal{H}(\mathbf{z}_1, -\mathbf{q}_1)$  then
8:     state  $\leftarrow 2$ 
9:   end if
10: else if state = 2 then
11:   if  $\mathbf{p} \in \mathcal{H}(\mathbf{z}_1, \mathbf{q}_1)$  then
12:     state  $\leftarrow 3$ 
13:   end if
14: else if state = 3 then
15:   flag  $\leftarrow 1, \mathbf{r} \leftarrow \mathbf{z}_1, \mathbf{q} \leftarrow \mathbf{q}_1$ 
16:   if  $\mathbf{p} \in \mathcal{H}(\mathbf{z}_2, \mathbf{q}_1)$  then
17:     state  $\leftarrow 4$ 
18:   end if
19: else if state = 4 then
20:   flag  $\leftarrow 2, \mathbf{c} \leftarrow \mathbf{c}_e, \rho \leftarrow R, \lambda \leftarrow \lambda_e$ 
21:   if  $\mathbf{p} \in \mathcal{H}(\mathbf{z}_3, -\mathbf{q}_3)$  then
22:     state  $\leftarrow 5$ 
23:   end if
24: else if state = 5 then
25:   if  $\mathbf{p} \in \mathcal{H}(\mathbf{z}_3, \mathbf{q}_3)$  then
26:     state  $\leftarrow 1$ 
27:      $i \leftarrow (i + 1)$  until  $i = N$ .
28:      $(L, \mathbf{c}_s, \lambda_s, \mathbf{c}_e, \lambda_e, \mathbf{z}_1, \mathbf{q}_1, \mathbf{z}_2, \mathbf{z}_3, \mathbf{q}_3) \leftarrow$ 
       findDubinsParameters( $\mathbf{w}_{i-1}, \chi_{i-1}, \mathbf{w}_i, \chi_i, R$ )
29:   end if
30: end if
31: return flag,  $\mathbf{r}, \mathbf{q}, \mathbf{c}, \rho, \lambda$ .

```

Dubins Path Following Results

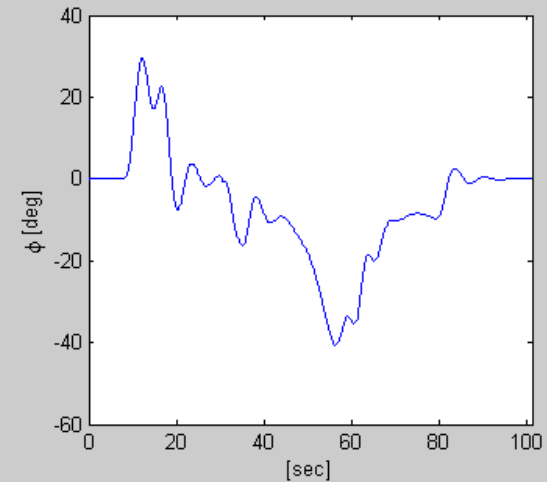
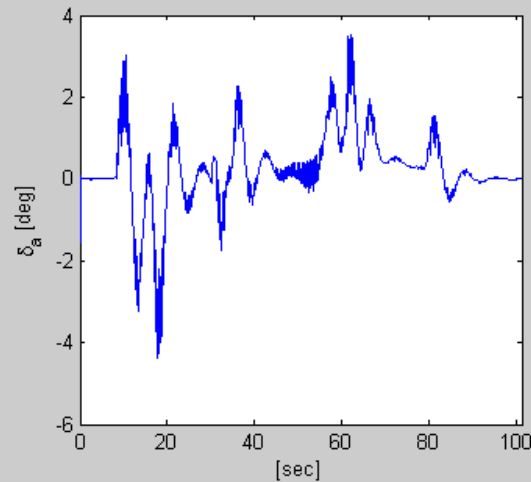
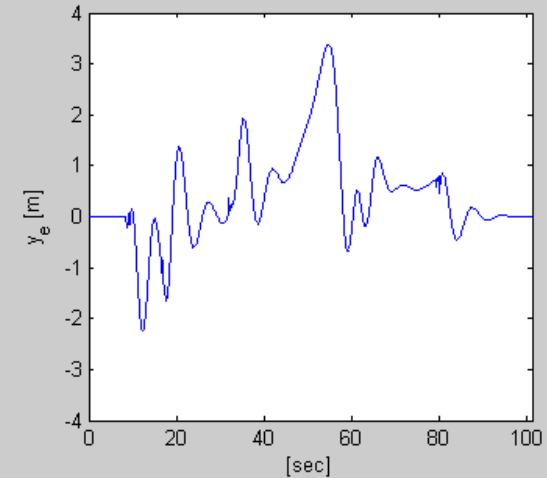
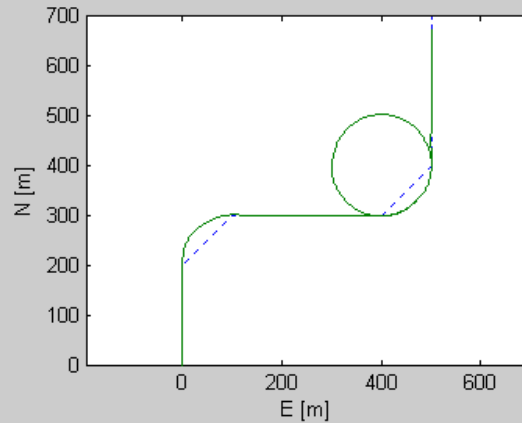


Other examples in simulation

Flying wing
(2.5kg, $b=2\text{m}$)

20m/s airspeed
Wind 8m/s

Min. Radius 100m
L1 guidance



Other examples in simulation

Quadrotor
2.9kg

Wind 7m/s
from North
w/ moderate
turbulence

Min.radius 20m

